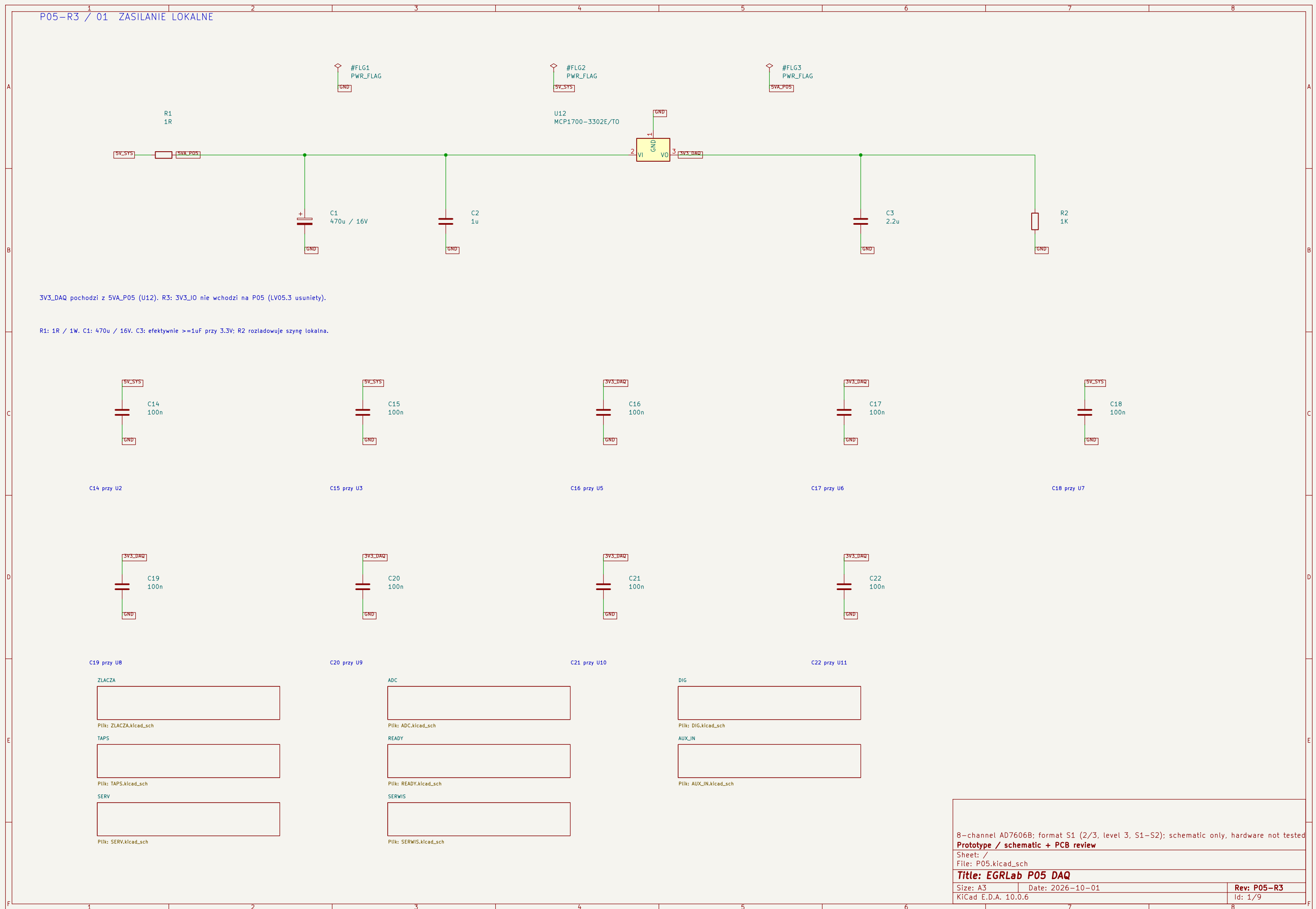




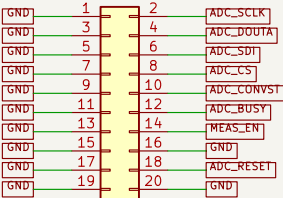
J_BP1 (slot S1) i J_BP2 (slot S2): katowe obudowane IDC na krawedzi A, tasma do P12. Pin 1 od strony mniejszego x.

J_BP2: DAQ na tych samych pinach co P03 R6 J_BP2. Piny 16 i 20 GND (rezerwa); 3V3_IO nie wchodzi na P05.

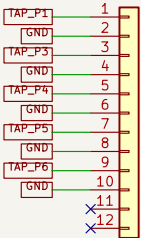
J_BP1
J_BP1 / IDC 2x5



J_BP2
J_BP2 / IDC 2x10

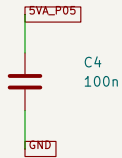
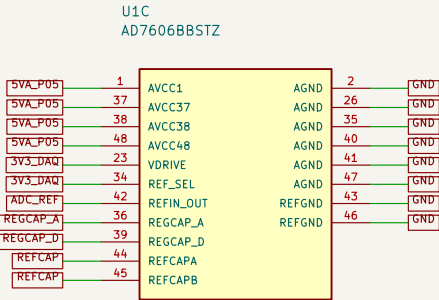
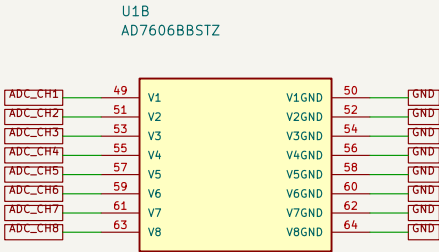
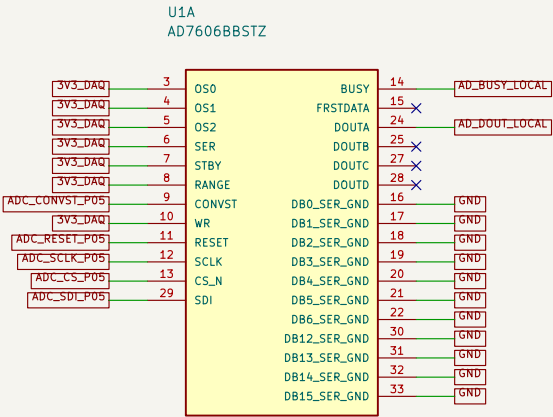


J4
TAPS / PTH

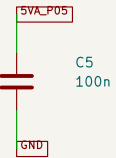


J4 TAPS: 50mm AWG24, 5 par, lutowane w PTH przy brzegu x=0 (S1: przewodem, nie przez P12).

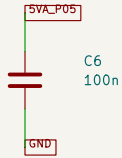
Rezystory 300k (motor) / 100k (sensor) pozostaja w adapterach AT/AL1/AL2 przy zrodle sygnalu.



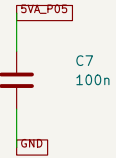
C4 przy U1.1



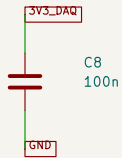
C5 przy U1.37/38



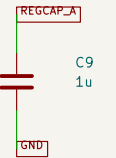
C6 przy U1 (dodatkowy 100n na SVA, miejsce w layoucie S1)



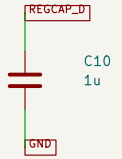
C7 przy U1.48



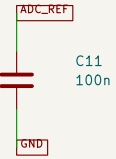
C8 przy U1.23



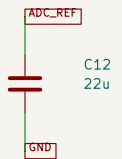
C9 przy U1.36



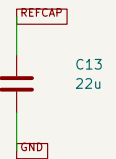
C10 przy U1.39



C11 przy U1.42



C12 przy U1.42



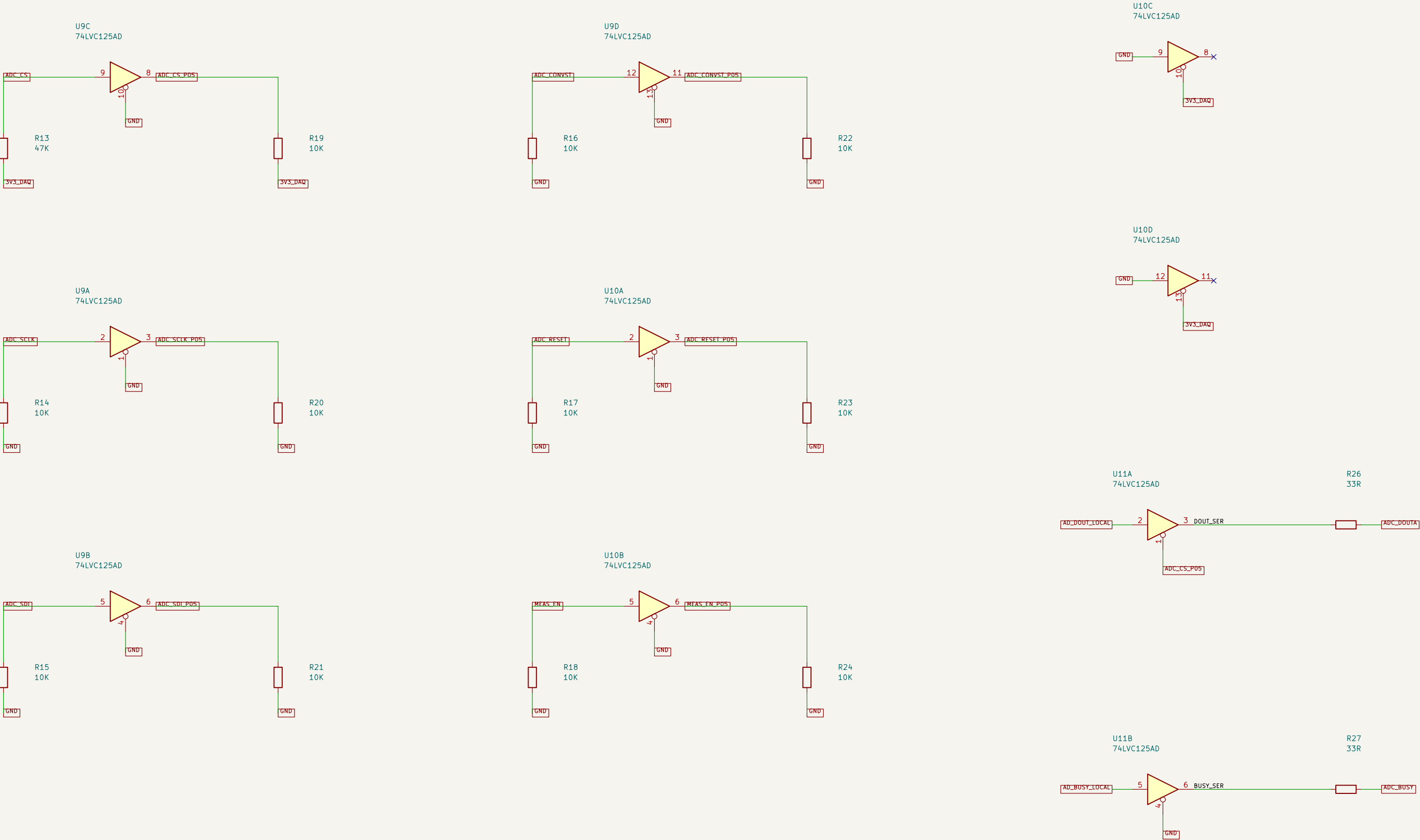
C13 przy U1.44/45

OS[2:0]=111; SER=1; WR10=1. RESET11: pełny reset >=3us. BUSY14 przechodzi przez U11B.

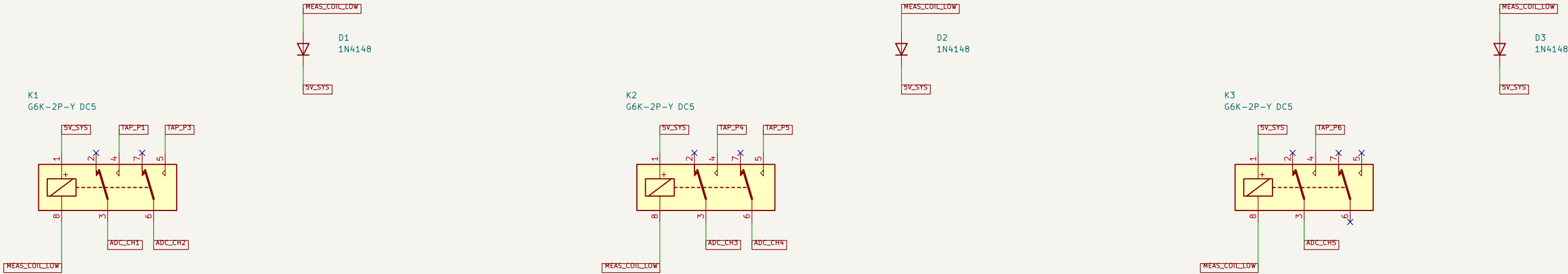
Referencja ADC wewnętrzna. C12/C13: 22uF/25V 1210 X7R, efektywnie >=10uF; 36/39 osobne 1uF.

Nie zwierać pinów 36/39. Wyjścia DOUTB/C/D i FRSTDATA pozostają NC. Serial DB0..6/12..15 = GND.

loff chroni wylaczona domene. U11A: OE_N=CS; wspolna linia MISO nie jest stale napedzana.

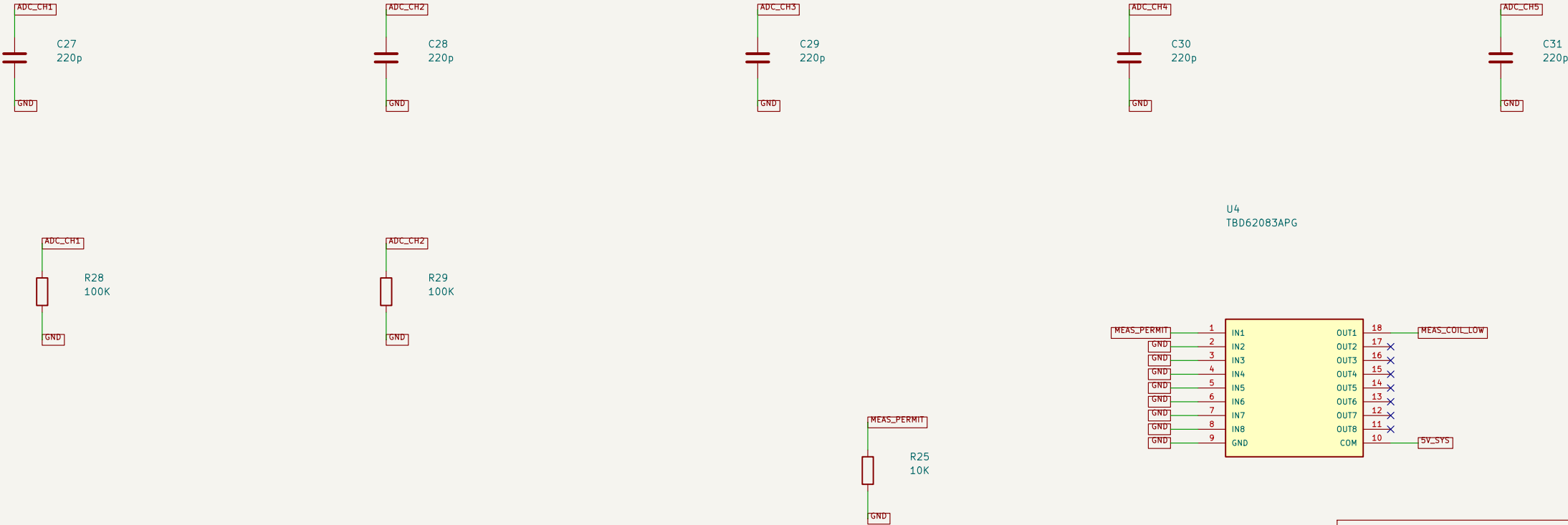


TAP: sygnał już za rezystorem w adapterze. Nie podłączać surowego motoru do J4!



K1/K2/K3 NO: otwarte bez zasilania. Złączenie tylko MEAS_EN & DAQ_OK; nie zależy od ARM.

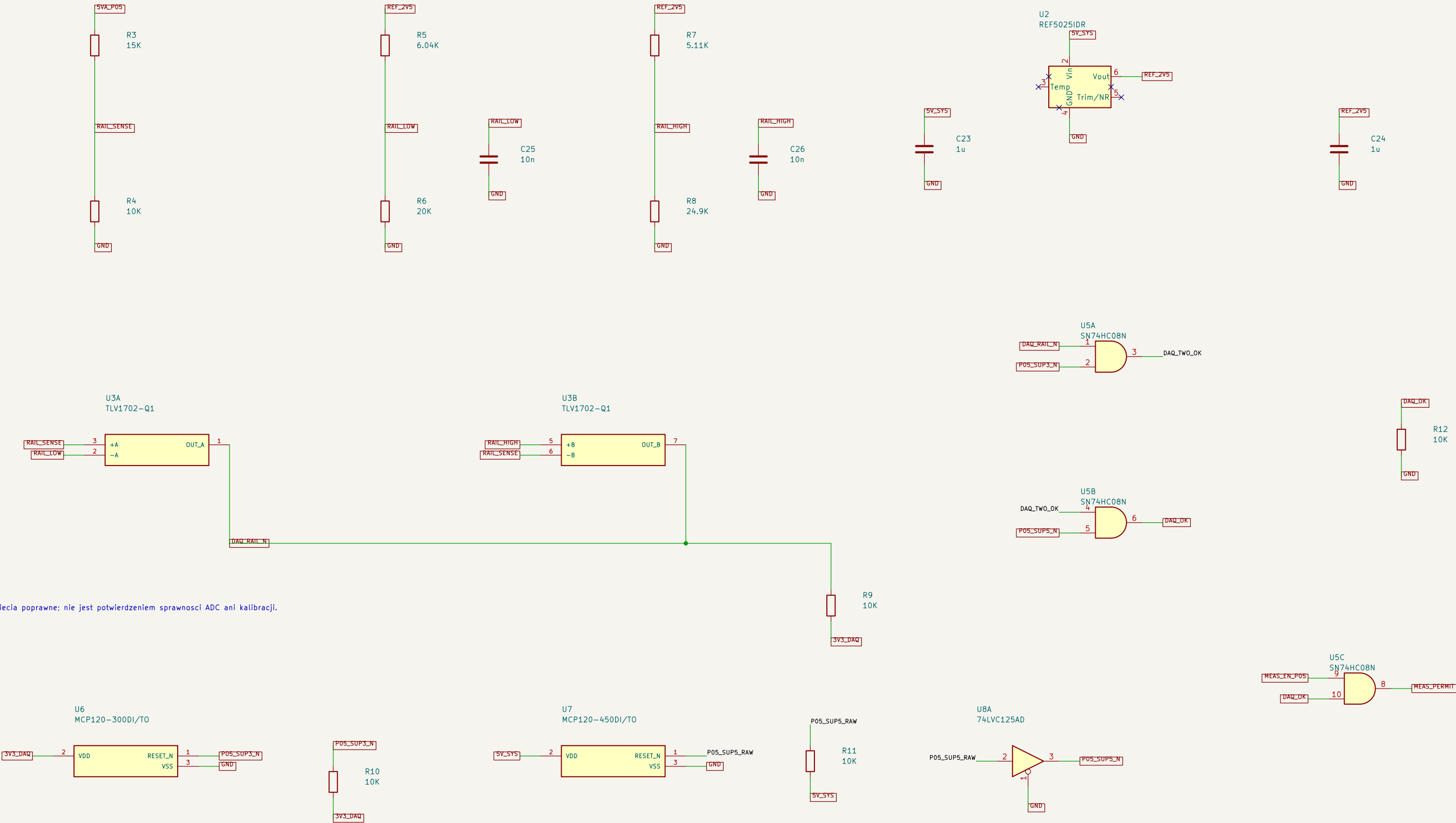
CH1/2: dolna gałąź 100k. CH3/4/5: bez 100k do masy (pomiar czujnika). Filtry COG 220p.



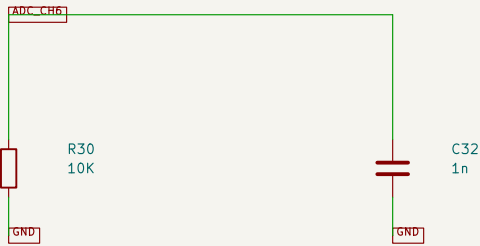
Okno nominalne 5VA: 4.800..5.186V (R5 6.04k, R7 5.11k). Tolerancje i VOS: verification/electrical-checks.json.

U2 REF5025IDR (klasa wysoka 0.05%, 3ppm/K). REF5025AIDR nie jest zamiennikiem: dolny naroznik < 4.75V.

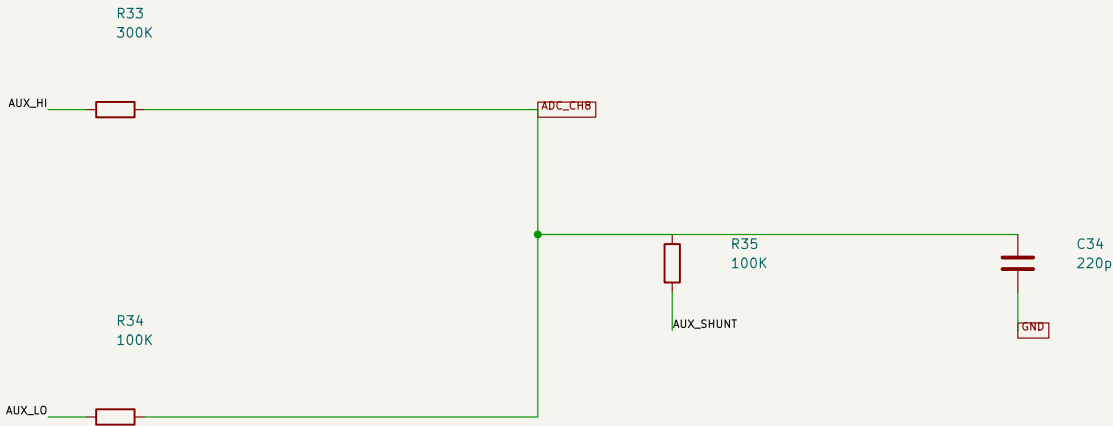
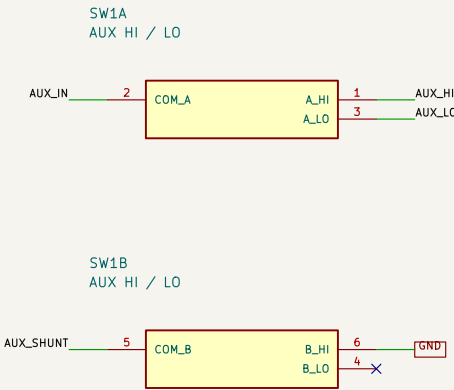
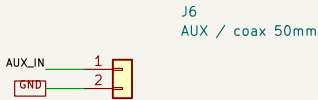
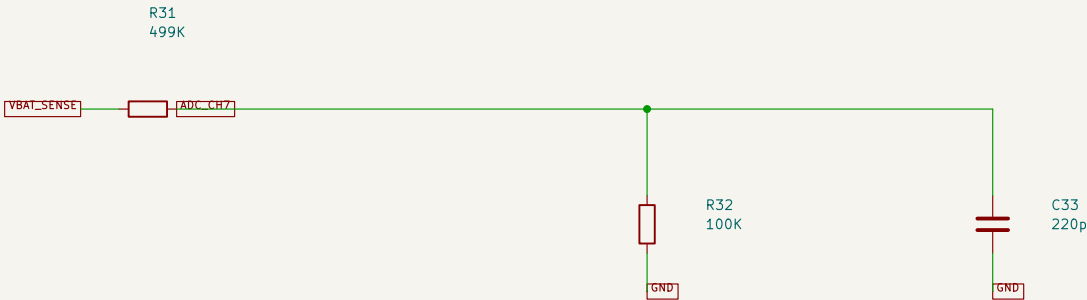
DAQ_OK: napiecia poprawne; nie jest potwierdzeniem sprawnosci ADC ani kalibracji.



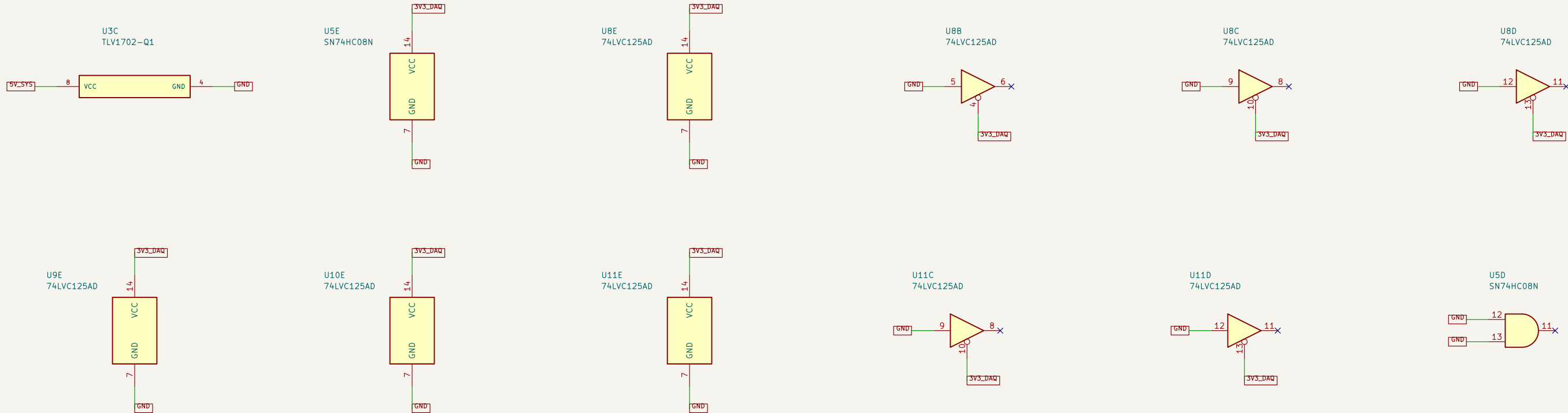
CH6: terminacja zera; prad z MCP3201 na P06/P07 jest osobnym pomiarem. Nie dodawac shunta do CH6.



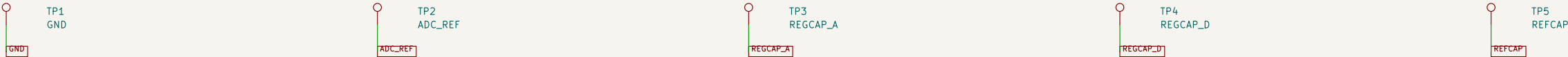
CH7 VBAT_SENSE (akumulator auta przez P02 R4 J11.1): 499k/100k, nominalny mnoznik 6.0898; kalibracja offsetu i gain obowiazkowa.



SW1 C&K JS202011AQN (suwak, katowy): HI = 2-1 + 5-6, LO = 2-3 + 5-4 (pin 4 wolny). Przelaczac bez napiecia; po zmianie profil AUX.



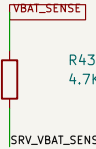
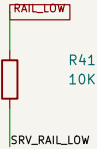
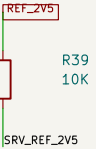
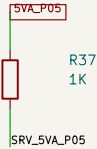
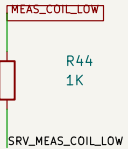
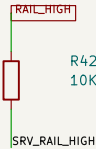
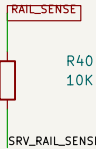
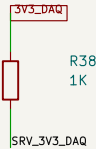
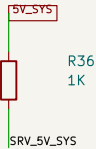
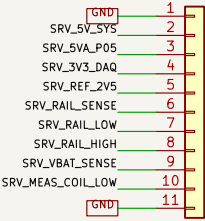
TP1..TP5 tylko przy U1 (krok 11 ODBIOR, przed skreceniem stosu): REF/REGCAP nie wychodza na listwe B (szum w referencji ADC).



J_SV1 (slot S1, x=10..43): zasilania i analog. J_SV2 (slot S2, x=63.5..96.5): sygnały DAQ. GND na pierwszym i ostatnim pinie.

Kazdy kolek przez rezystor przy wezle: 1k szyny/logika, 4.7k VBAT_SENSE, 10k wezly wysokoimpedancyjne. Numeracja zawsze od pinu 1.

J_SV1
J_SV1 SERWIS 1x11



J_SV2
J_SV2 SERWIS 1x13

